

open source in enterprises, and is getting reflected in CIO and CTO interactions too.

Q. How is Linux growing in terms of its share on the Power Systems platform with the introduction of SAP HANA kind of applications?

A. Linux has been there for quite some time (approximately 20 years) on the power systems. Back then it was running on the IBM Hypervisor named PowerVM in a virtualised environment. In 2014, Linux on IBM Power Systems spread out from its high-performance computing domain to other areas with a focus on the cloud, AI and enterprise databases like SAP HANA. And that trend is sort of continuing with the growth of our business.

The same holds true with respect to the mainframe, System Z, which is running Linux as well. A lot of incredible technologies have been brought onto the System Z platform for scaling to hundreds of billions of transactions and providing end-to-end security. So, considerable progress has been made on both these platforms in terms of adopting and optimising Linux. Compared to the Linux market's growth, IBM's Linux business is growing at a faster rate.

Q. Within your IBM systems (both Z and Power Systems platforms), is the share of Linux servers increasing vis-a-vis your other solutions?

A. On a year-to-year basis, our Linux business is growing considerably well at the moment. This growth has been a part of our long-term strategy. Regarding the IBM Power series of servers running Linux, we have had a presence in the HPC market in the past where a lot of top performing

HPC systems have been IBM power systems. About six years ago, a number of our partners, like Mellanox, Google and Rackspace from the OpenPOWER Foundation, started collaborating with us on the possibility that these systems could also be very good for the cloud and AI market. And on discovering new markets and new possibilities related to AI and the cloud, we directed our efforts on these new technologies and built a successful product line.

Q. According to you, what are the key technology trends that drive the evolution of Linux and keep it relevant for the future?

A. Linux's advantage has been that it's so configurable that it can be run on pretty much any kind of platform, all the way from embedded systems to very large enterprise-class systems.

We're seeing Linux actually being used not just in the main system, but also running on many other parts of a server system, like the embedded controllers that manage a server system and I/O adapters.

It is well known by now that reliability and high-availability technologies in Linux are very useful for high-end enterprise software and solutions. But there is also work going on to make Linux a very good player for the cloud.

Memory technologies are a good example of new innovations coming to Linux. Memory is no longer limited to just the DRAM technology. Within the system memory, we now have different classes of memory technologies, e.g., DRAM, storage class memory, etc — all of which have different performance characteristics and different semantic behaviour. So, a lot of work is happening in Linux right now, to handle multiple memory technologies in a single system.

So, I see this trend of a lot of new Linux innovations emerging that are based on new system technologies, be it on the processor architecture side or the memory architecture side.

Q. Could you tell us a bit more about the OpenPOWER Foundation?

A. The OpenPOWER Foundation is a global organisation comprising like-minded industry and academic partners. It includes people who work on hardware, chip design, system design and system engineering, all the way up to cloud software, high-performance computing software and academic research. Currently, it has over 300 members.

The OpenPOWER Foundation began as a small organisation initiated by partners like IBM, Mellanox, NVIDIA and Google. The organisation slowly expanded to include a large pool of hardware and software vendors. Some of the partners, like Suzhou PowerCore from China, belong to the Asia-Pacific region.

Q. Does the organisation operate in India?

A. OpenPOWER does have academic partners from several universities in India, who are working on research projects around the OpenPOWER platform. For example, we have a research collaboration with one of the OpenPOWER Foundation academic members in India, who is from the National Institute of Technology, Surathkal, Karnataka. They are working on the popular open source architecture simulator called gem5, and adding power processor and system architecture support in it. For an academic research group to have such a modern architecture simulator, which can do performance evaluations, is very important. The OpenPOWER Foundation is quite supportive of this work as well. 